

Object Identification for Sequence Diagram based on Use Case Scenario using Text Semantic: Case Study of IdVar4CL Artefact

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Abstract— *Inconsistencies in the creation of Unified Modeling Language (UML) artifacts occur due to differences in interpretation by each individual. These differences may arise because each individual has a different understanding of the causes and effects in an environment/system, influenced by experience, learning process, insight, intuition, and assumptions. In the process of identifying variables in a case study called IdVar4VL, there is an inconsistency between the artifacts, making this case an example of implementation in identifying UML artifacts for conformity. Our research aims to identify objects derived from all Use Case Scenario tabulations, and then use the resulting objects for the Sequence Diagram. This research has implemented stages to identify objects that produce datasets in the form of Use Case Diagrams consisting of 3 actors. The Use Case Scenario tabulation produces 10 Step-Perform documents, namely d2 to d11. After all stages are completed, this research produces an alternative dataset for UCD in the form of OCR text document, forming object tabulations based on 10 Step-Performs from 10 UCS and making improvement recommendations for Step-Performs in UCS artifacts. Additionally, the validity and reliability of the generated object tabulation against the Step-Performed score is 0.23, calculated using Gwet's ACI formula. The resulting scores fall into the "Fair Agreement" category based on Gwet's ACI index, making the generated object tabulation a reference for forming a complete Sequence Diagram.*

Keywords— *Object, Sequence Diagram, Text Semantic, Use Case Diagram, Use Case Scenario, UML.*

I. INTRODUCTION

In the formation of Unified Modeling Language (UML) artifacts, there are inconsistencies with other artifacts [1]. This inconsistency occurs due to differences in interpretation in each individual, where this difference can occur because each individual has a different understanding of the causes and effects that occur in an environment/system, such as experience, learning process, insight, intuition, and assumptions [2]. For example, making Sequence Diagrams manually is very error-prone and takes a lot of energy and time [3][4]. This error can happen due to Use Case writing that uses Natural Language. Besides making the Use Case to be easy to understand by stakeholders, it also makes the Use Case incomplete and ambiguous [5][6]. This deficiency makes the resulting Sequence Diagram not a direct representation of the existing Use Case [7]. Therefore, it is necessary to conduct an in-depth analysis and ensure the integration of the Use Case with other Unified Modeling Language (UML) artifacts, especially for the sake of documentation and software development [6][8]. The creation of Sequence Diagrams can also be biased and inconsistent if a new added message, adding one new object that is the same as an existing object in

the Sequence Diagram [9]. To overcome this problem, it is necessary to analyze the Use Case to form relevant activities and identify the objects involved in these activities [5].

Through Text Pre-Processing and Text Semantic activities, Use Case data can be processed to produce a semantically equivalent set of objects that is better in understandability and maintainability to help develop other Unified Modeling Language (UML) artifacts [10]. A semantically equivalent set of objects can be used to form Sequence Diagrams that are better in terms of accuracy and completeness than manually generated Sequence Diagrams [11]. These activities can also save enormous costs and effort in developing and improving software and help non-developers understand the behavior of the system [3][4][7].

The case study used in our research is called Identification Variable for Causal Loop (IdVar4CL). In this case study, there is a data extraction process on the storytelling description of a systems thinking case through text mining and similarity that can identify variables to form causal loop diagrams. In the process of identifying variables for this case study, inconsistencies were found between the artifacts, so this case can be used as an example of implementation in identifying Unified Modeling Language (UML) artifacts so that there is conformity [2].

This research aims to identify objects derived from all Use Case Scenario tabulations, which will then be used on objects in the Sequence Diagram. To achieve this goal, some activities will be carried out, namely:

- Prepare dataset that consist of all Use Case Diagram artifacts and Use Case Scenario tabulations contained in the Idvar4CL (Software Requirements Specifications) SRS documentation.
- Perform Text Pre-Processing activities on all research-related artifacts, namely, Use Case Diagram and Use Case Scenario.
- Processing data to find semantic similarity between object-forming artifacts for Sequence Diagrams.
- Perform validity and reliability testing on the results of object identification used in the Sequence Diagram.

II. RELATED WORK

A. Identification Variable for Causal Loop (IdVar4CL)

There is research that has been conducted and explored on the use of data mining to complete Software Requirement

Specification (SRS) documentation on the IdVar4CL case study. Priyadi et al. [2] used a text mining approach to generate variable approaches in the form of groups of words, which will be used in modeling system thinking based on the value of connectedness greater than 0.50. Through the same case study, this research helps to improve the quality of IdVar4CL application development. In addition, some studies use text semantics methods to generate Unified Modeling Language (UML) artifacts in IdVar4CL application development, namely sprint stages for developing applications, Functional Requirement—Non-Functional Requirement (FR-NFR), and Use Case Scenario (UCS) [12][13][14].

B. UML Artefacts: Use Case Scenario (UCS) and Sequence Diagram (SD)

Research conducted by Sharp et al. [9] and Briand et al. [7] used the Reverse Engineering approach in forming a Sequence Diagram. Then there is research conducted by Thakur et al. [11], and Alami et al. [3], who formed Sequence Diagram (SD) using a toolkit in the form of MADA + TOKAN NLP Tool and Natural Language Parser Model. Then, the research was conducted by Lu et al. [10], and Canal et al. [15], who have implemented the Multi-Objective Search Algorithm and Levenshtein Algorithm. Furthermore, research conducted by Hamza et al. [16], Sawprakhon et al. [4], and Ellaloui et al. [1], who have implemented a Model-Driven Approach in the form of a Use Case Analysis Model, Atlas Transform Language (ATL), and scrum integrated Model Driven Architecture. Finally, research conducted by Almendroz et al. [8] and Yang et al. [17] who have implemented Semantic Text and a developed tool, namely Requirements Modeling and Automatic Prototyping (RM2PT) to create Use Cases and explain the Use Case relationship with Sequence Diagrams. All these studies have a significant impact on the completeness of Software Requirement Specification (SRS) documentation, especially in terms of Use Case formation and Sequence Diagram formation.

C. Text Processing

There is research that has explored the concept of Natural Language Processing (NLP) and the implementation of Text Processing. According to Ding et al. [5] and Fatwanto [6], NLP is a concept that focuses on the ability of computers to understand, interpret, and generate human language. Then, research by Seba et al. [12] and Schwitter [18], have implemented Text Pre-Processing (TPP) and converted datasets into a structured form to produce Unified Modeling Language (UML) artifacts in the form of application development sprint stages and objects for Sequence Diagrams. Finally, research by Fatimatuzzahra et al. [13] and Laksana et al. [14], implemented Text Semantic to find semantic conformity from the dataset and form Unified Modeling Language (UML) artifacts in the form of Functional Requirement—Non-Functional Requirement (FR-NFR) and Use Case Scenario (UCS). Through the implementation of Text Processing, this research helped to improve the information quality of Unified Modeling Language (UML) artifacts for IdVar4CL application development.

D. Validity and Reliability Testing

There are studies that have implemented Gwet's AC1 and Cohen's Cappa formulas to test the validity and reliability of

research results. Gwet's AC1 is a testing formula that uses different agreement-chance models to increase the stability of the dataset. At the same time, Cohen's Cappa is a testing formula that measures the comparison between the tested agreement and other random agreements. Research conducted by Jimenez et al. [19] Ohyama [20] and Cibulka et al. [21] used Gwet's AC1 formula and got the final product category of the research results "Slight to Moderate Agreement". On the other hand, research conducted by Schwitter [18], Werner et al. [22], and Primadiati et al. [23] have used the Cohen's Cappa formula and obtained the final product category of research results "Fair to Substantial Agreement".

Through this implementation, these research studies helped improve the accuracy of the IdVar4CL Software Requirement Specification (SRS) documentation validation process. Gwet's AC1 formula and Cohen's Cappa index table are attached below:

- Forming the Elicitation Statement table between two observers, attached as Table I.

TABLE I. ELICITATION STATEMENT RESULT

Expert 1	Expert 2		
	Yes	No	Total
Yes	A	B	B1 = A + B
No	C	D	B2 = C + D
Total	A1 = A + C	A2 = B + D	N

- Calculating the Agreement Coefisien through the AC1 formula.

$$P = \frac{A+D}{N} \quad (1)$$

- Calculating the odds-agreement probability.

$$e(\gamma) = 2P_1(1 - P_1) \quad (2)$$

P_1 can be calculate using the following formula:

$$P_1 = \frac{(A_1 + B_1)/2}{N} \quad (3)$$

A_1 and B_1 are Observer 1 and Observer 2, and N is the total amount of data.

- Classifying the probability result data into "Yes" categories.

- Calculating AC1 statistics.

$$AC1 = \frac{P - e(\gamma)}{1 - e(\gamma)} \quad (4)$$

- Comparing the statistical results with the agreement index between two observers.

- Classifying the final results of the calculation, as attached in Table II.

TABLE II. INDEX VALUE COHEN'S CAPPA

Index	Agreement Proportion
< 0	Less than Chance
0.01 – 0.20	Slight
0.21 – 0.40	Fair
0.41 – 0.60	Moderate
0.61 – 0.80	Substantial
0.81 – 1.00	Almost Perfect

E. Optical Character Recognition (OCR)

Optical Character Recognition (OCR) is a technique that is often used to extract text from an image. There is research that has explored the concept of Optical Character Recognition (OCR) for character recognition, especially on

vehicle license plates. Bansal et al. [24] used the Optical Character Recognition (OCR) method to capture and process vehicle license plate images to read vehicle license plates and check with document management systems. Through the implementation of the Optical Character Recognition (OCR) method, this research offers a method to provide alternative datasets in the form of text to facilitate data processing from datasets in the form of images.

III. DATASETS

This research uses 11 datasets to process data. This dataset consists of 10 Use Case Scenarios and 1 Use Case Diagram. For the Use Case Diagram (UCD) dataset, the diagram is converted into text form firstly. The Use Case Diagram text document is obtained using the Optical Character Recognition (OCR) technique (See Fig. 1). Then, for the Use Case Scenario (UCS) dataset, it will take each Step-Perform that has a Subject-Verb-Object (SVO) structure from each Use Case Scenario (See Fig. 2).

Title: IdVar4CL Application. Actors: Researcher, Admin, IdV Login, Admin Login, Modify Application Settings, Input mult: Normalize Text, Generate Variable Set, Calculate Cosine Sim: Highest Similarity. Relationships: <<include>>, <<extend>>.

Fig. 1. Snippet of Text Document from UCD's OCR

Step performed: Researchers and admins open the IdVar4CL app: username and password on the login page. The IdVar4CL applic: data in the database. If valid, the application displays a di: admin). If it is invalid, the application displays an error: the login process.
Step performed: After successful login, the IdVar4CL applica: (Researcher or admin). If it has been determined, they will: role, access to additional features, such as application set: If the user is a researcher, the available access is only fo: settings.
Step performed: Researchers enter the document input page. R: their toolkit. The researcher clicks the 'upload' button to: The IdVar4CL application validates the document format and c:

Fig. 2. Snippet of Step-Perform from UCS

IV. METHODOLOGY

Referring to Fig. 3, an attached diagram illustrates the flow of the research conducted. The following is an explanation of each stage:

1) Document Analysis

This stage aims to prepare initial data in the form of artifacts from the Use Case Diagram (UCD) and Use Case Scenario (UCS). The Use Case Diagram (UCD) is processed using Optical Character Recognition (OCR) techniques to convert each element of the diagram into text. Then the Use Case Scenario (UCS) is analyzed based on Step-Performed, which is the steps or activities performed in the scenario. This process produces raw data that will be used in the data processing and application development stages.

2) Application Development

This stage consists of 3 main steps, namely:

- **Extraction:** Extracting Optical Character Recognition (OCR) data from Use Case Diagram (UCD) and Step-Perform from Use Case Scenario (UCS) for data analysis and processing.

- **Text Pre-Processing:** preparing and cleaning the dataset into a structured form using Case Folding techniques that convert each letter in the text into lowercase letters, Tokenizing techniques that break the text into small groups, Stemming techniques that return words to their basic form, and Stop-Word Removing techniques that remove common words without meaning.
- **Text Semantic:** Calculates the semantic similarity of each Use Case Diagram (UCD) and Use Case Scenario (UCS) element using word weighting through Term Frequency—Inverse Document Frequency and machine learning model through Word2vec. The semantic calculation produces a semantic value that represents the relationship between each Use Case Diagram (UCD) and Use Case Scenario (UCS) document.

3) Validity and Reliability Testing

This stage aims to evaluate the results of object identification that has been carried out. Validity testing uses Gwet's AC1 method to measure the level of agreement between observers with more stability on datasets with unbalanced category distribution. In addition to testing validity, reliability testing is also carried out through a questionnaire of researchers as an Expert Questionnaire to ensure conformity between the objects produced and the opinions of experts.

4) Final Result

The end result of this entire process is a tabulation containing a list of identified objects for use in creating Sequence Diagrams. This tabulation includes the relationships between the Use Case Scenario (UCS) steps and the objects in the Use Case Diagram (UCD), which have been validated and tested for reliability.

Referring to all the explanations above, this research will complete one of the Unified Modeling Language (UML) artifacts in the IdVar4CL Software Requirement Specification (SRS) documentation, namely the Sequence Diagram. More specifically, this study tabulates objects for Sequence Diagram that are semantically similar based on Use Case Scenario data using Text Processing methods, namely Text Pre-Processing and Text Semantics.

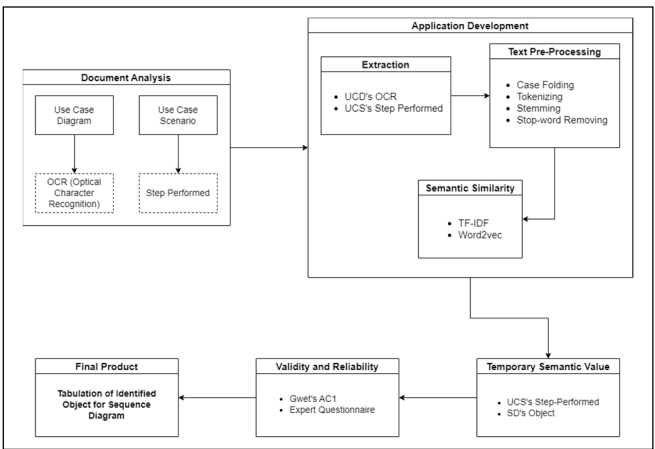


Fig. 3. Methodology

V. RESULT AND DISCUSSION

A. Document Analysis

At this stage, a document analysis was conducted to obtain the required dataset. For the Use Case Diagram (UCD) document, an Image-to-Text conversion using the Optical Character Recognition (OCR) method is performed, and produces a text document containing an explanation of the elements of the Use Case Diagram (UCD). The Optical Character Recognition (OCR) document text from the Use Case Diagram (UCD) is labeled d1 and stored in a file named "ucd_ocr.txt". Then, for the Use Case Scenario (UCS) document, Step-Perform extraction of each Use Case Scenario (UCS) from the IdVar4CL artifact is performed. The extracted document text is labeled d2-d11 and stored in a file named "ucs_st.txt". Fig. 4 presented the results from data labeling.

```
#UCD Document
d1 = "title: idvar4cl application. actors: rese
ettings, input multiple text document, authenti
l loop diagram, display highest similarity. rel

#Step-Perform Document
d2 = "step performed: researchers and admin
he idvar4cl application verifies the user na
role (researcher or admin). if it is invali
d3 = "step performed: after successful logi
determined, they will be redirected to the
ble if the user is an admin. if the user is
```

Fig. 4. Snippet of Data Labelling

B. Text Pre-Processing

In this stage, dataset cleaning is done to prepare each dataset for the next stage. Cleaning is done by Case Folding (changing each letter in the text to lowercase), Tokenizing (breaking the text into small groups), Stemming (returning words to their basic form), and Stop-Word Removing (removing common words without meaning). After cleaning, the Use Case Diagram (UCD) dataset is saved in a new file named "ucd.txt" and the Use Case Scenario (UCS) dataset is saved in a new file named "st.txt". Fig. 5 presented the results from Text Pre-Processing.

```
Stemmed UCD Text Documents:
titl : idvar4cl applic . actor : research , admin , idv.
text document , authent , process text data , normal te:
lar . relationship : < < includ > > , < < extend > > .

Stemmed Step-Perform Text Documents:
step perform : research admin open idvar4cl applic . research
bas . valid , applic display dashboard accord role ( research
step perform : success login , idvar4cl applic check databas
featur , applic set , avail user admin . user research , avai
step perform : research enter document input page . research
pplic . idvar4cl applic valid document format save comparisor
```

Fig. 5. Snippet of Pre-Processing Dataset

C. Document Weighting

At this stage, text weighting is carried out using the Term Frequency—Inverse Document Frequency (TF-IDF) method. This text weighting aims to produce a value or score that represents how important each word in each document is. The

results of text weighting from each document can be seen in Fig. 6.

UCD TF-IDF Representation:									
	actor	admin	applic	authent	calcul	case	causal	\	
0	0.116248	0.232495	0.348743	0.116248	0.116248	0.116248	0.116248		
ST TF-IDF Representation:									
	accept	access	accord	addit	admin	algorithm	allc		
0	0.000000	0.000000	0.127948	0.000000	0.398888	0.000000	0.000000		
1	0.000000	0.273506	0.136753	0.136753	0.212742	0.000000	0.000000		
2	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000		
3	0.161228	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000		
4	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000		
5	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000		
6	0.000000	0.000000	0.000000	0.000000	0.099533	0.000000	0.150571		
7	0.000000	0.165276	0.000000	0.165276	0.385672	0.194422	0.000000		
8	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000		
9	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000		

Fig. 6. Snippet of Weighting Document

D. Similarity Process

In this stage, semantic congruence is calculated using the Text Semantic method, which involves the Word2vec model. The Word2vec model calculates semantic congruence by converting the words in each text into vectors and calculating each vector to produce a value that represents the relationship of each word in each document. The calculated value of semantic congruence can be seen in Fig. 7.

Similarity Matrix:				
	UCS Document 1	UCS Document 2	UCS Document 3	\
UCD Document 1	0.831745	0.725537	0.739959	
	UCS Document 4	UCS Document 5	UCS Document 6	\
UCD Document 1	0.710511	0.723099	0.736412	
	UCS Document 7	UCS Document 8	UCS Document 9	\
UCD Document 1	0.756132	0.793444	0.534561	
	UCS Document 10			
UCD Document 1	0.625453			

Fig. 7. Snippet of Semantic Similarity

E. Validity and Reliability Testing

At this stage, the validity and reliability testing process is carried out on the results of calculating the value of semantic conformity. Testing is carried out using an expert questionnaire that can be accessed via <https://forms.gle/wppx3LEmWH935VzX7>. Respondents who filled out the questionnaire were researchers and software consultant in the field of software engineering from universities. The results of filling out the questionnaire can be seen in Table III.

TABLE III. QUESTIONNAIRE RESULT

Expert 1	Expert 2		
	Yes	No	Total
Yes	8	2	10
No	6	4	10
Total	14	6	20

Based on the questionnaire results, validity and reliability testing is then carried out using Gwet's AC1 formula, as follows:

$$P = \frac{8 + 4}{20} = 0.6 \quad (5)$$

$$P_1 = \frac{(14 + 10)/2}{20} = 0.6 \quad (6)$$

$$e(\gamma) = 2(0.6)(1 - 0.6) = 0.48 \quad (7)$$

From the calculation results above, the Gwet's AC1 value obtained is as follows:

$$AC1 = \frac{0.6 - 0.48}{1 - 0.48} = 0.23 \quad (8)$$

The result of the validity and reliability calculation is 0.23. Based on these results, the test results fall into the "Fair Agreement" category. Based on the results of this category, the score or value of the test results shows that the final product produced is able to be a reference in forming Sequence Diagrams in future research. The object tabulation for Sequence Diagram based on Use Case Scenario can be seen in Table IV. Then, for Step-Perform improvement, recommendations on the Use Case Scenario artifacts can be seen in Table V.

TABLE IV. OBJECT TABULATION

No.	Use Case Scenario	
	ID	Object in Step-Perform
1.	UCS01	IdVar4CL Application, Login Page, Dashboard Page
2.	UCS02	Database, Dashboard Page
3.	UCS03	Document Input Page, Text Document
4.	UCS04	Text Document
5.	UCS05	Text Document, Similarity Score
6.	UCS06	Text Document, Variable Set
7.	UCS07	Dataset, Causal Loop Diagram
8.	UCS08	Application Setting, Algorithm Parameters, External Tool
9.	UCS09	Similarity Score, Text Document
10.	UCS10	Text Document, Normalized Text Document

TABLE V. STEP-PERFORM IMPROVEMENT RECOMMENDATION

No.	Use Case Scenario	
	ID	Statement Step-Perform
1.	UCS01	The user (researcher(s) or admin(s)) opens the IdVar4CL application. The user (researcher(s) or admin(s)) enters their username and password on the login page. The application verifies the user's username and password with data in the database. The application displays the dashboard based on the user role (researcher or admin) if valid. The IdVar4CL application displays an error message and asks the user to repeat the login process if invalid.
2.	UCS02	The application checks the database to determine the user role (researcher or admin). The user is redirected to the dashboard based on the user role.
3.	UCS03	The user (researcher(s)) enters the document input page. The user (researcher(s)) selects the text document(s) from their toolkit. The user (researcher(s)) confirms the selected text document by clicking the "upload" button to add the selected text document into the application. The application validates the text document format and saves it for comparison.
4.	UCS04	The application accepts the uploaded text document. The application performs a case folding process to change all letters to lowercase. The application performs a tokenizing process to break down the text into words. The application performs a stop-word removing process to remove unimportant words. The application performs a stemming process to convert words into elemental forms.
5.	UCS05	The application converts each of the text documents to a vector after text preprocessing. The application calculates each of the text document's vector's angle using the cosine similarity method. The application

		processes a pair of the text documents to produce a similarity score between 0 and 1.
6.	UCS06	The application processes the text document to identify essential keywords after text analysis has been conducted. The application evaluates a relationship between essential keywords by calculating the connectedness value. The application adds an essential keyword to the variable set if the connectedness value is above 0.5. The application saves the variables set for use in causal loop diagram visualization.
7.	UCS07	The application receives a request to perform an analysis on the selected dataset. The application processes the selected dataset to identify the relationship between variables. The application generates a causal loop diagram based on the analysis result. The application displays the causal loop diagram in visual format. The user (researcher(s) and admin(s)) visually interprets a relationship between variables based on the causal loop diagram.
8.	UCS08	The user (admin(s)) accesses the application's setting feature after successfully logging in. The user (admin(s)) modifies the text mining algorithm parameters (connectedness or stemming limits). The user (admin(s)) integrates external tools for additional analysis. The application saves and implements the new setting after modification.
9.	UCS09	The application arranges the document's cosine similarity score from the highest to the lowest. The application displays the highest document's cosine similarity score at the top.
10.	UCS10	The application performs a lemmatization process to change the document's text to its basic form. The application performs an index process to record the normalized document's text. The application saves the normalized document's text for subsequent processes, such as searching or comparing to the other document.

VI. CONCLUSION AND FUTURE WORKS

This research has implemented stages to identify objects that produce datasets in the form of Use Case Diagrams consisting of 3 actors, namely: Researcher, Admin, and IdVar4CL Application. The Use Case Scenario Tabulation produces 10 Step-Perform documents, namely d2 to d11.

After all stages have been successfully executed, this research produces an alternative dataset for Use Case Diagram (UCD) in the form of Optical Character Recognition (OCR) document text, which has formed object tabulations based on 10 Step-Performs from 10 Use Case Scenario (UCS), and has made improvement recommendations for Step-Performs in Use Case Scenario (UCS) artifacts. In addition, there is validity and reliability of the object tabulation generated against the Step-Performed, so the score is 0.23. This score was obtained by calculating the questionnaire results using Gwet's AC1 formula. The resulting validity and reliability scores fall into the "Fair Agreement" category based on Gwet's AC1 Index. Based on this category, the object tabulation that has been produced can be used as a reference to form a complete Sequence Diagram.

Future work will involve several things, such as text dataset testing and assessment, improving the text in the dataset by following subject-verb-object sentence structure, providing alternative datasets for image-type datasets by implementing the Optical Character Recognition (OCR) method through any kind of tool, and lastly, automatically forming objects for the Sequence Diagram.

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